PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: Bee2Bee: Autonomous B2B Restocking Settlement Copilot

Version: 2.0

1. Project Overview

- **Problem Statement:** Small and medium-sized enterprises (SMEs) often face significant inefficiencies in managing procurement workflows. Inventory monitoring, supplier communication, invoice verification, and payment preparation are typically handled manually across fragmented channels such as emails, chat messages, PDFs, and spreadsheets. This results in delayed restocking, inconsistent decision-making, increased operational costs, and a higher risk of human error, particularly when dealing with unstructured supplier inputs and cross-border transactions.
- **Target Domain:**The system targets AI-powered procurement automation for SMEs, particularly in retail, wholesale, and e-commerce environments where inventory turnover and supplier coordination are frequent and critical operations.
- **Proposed Solution Summary:** Bee2Bee is an AI-powered, stateful procurement workflow system that transforms fragmented and unstructured inputs into structured, actionable workflows. It continuously monitors inventory, analyzes historical sales data, dynamically determines restocking needs, and autonomously manages supplier communication and negotiation. The system further processes invoices from unstructured formats, validates them against negotiated terms, and generates payment-ready outputs, effectively shifting the user's role from manual executor to strategic overseer.

2. Background & Business Objective

- **Background of the Problem:** Traditional procurement processes follow a manual and linear workflow, where inventory decisions are based on fixed thresholds rather than dynamic demand, and supplier communication occurs through informal and unstructured

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channels. This makes it difficult to extract meaningful data for automation and often leads to inefficiencies in negotiation, restocking, and financial reconciliation. The lack of integration across different stages of procurement further complicates operations and limits scalability for growing businesses.

- **Importance of Solving This Issue:** Addressing these inefficiencies is critical for improving operational speed, reducing costs, and minimizing risks associated with human error. By automating procurement workflows, businesses can respond more effectively to demand fluctuations, optimize supplier interactions, and ensure financial accuracy. This ultimately enhances business agility and competitiveness.
- **Strategic Fit / Impact:** The solution aligns with the growing adoption of AI-driven automation in enterprise workflows, demonstrating how large language models can serve as a central reasoning engine for real-world business operations. It showcases the potential of integrating AI with workflow orchestration systems to deliver end-to-end automation, which is highly relevant in today's digital transformation landscape.

3. Product Purpose

3.1. Main Goal of the System: The main goal of Bee2Bee is to provide an AI-enabled procurement copilot that automates the entire restocking lifecycle, from detecting low inventory to negotiating with suppliers, validating invoices, and preparing payments, while maintaining human oversight for critical decisions.

3.2. Intended Users (Target Audience):

- The system is designed for SME owners, procurement managers, inventory managers, and e-commerce merchants who require efficient and scalable solutions for managing inventory and supplier relationships.

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4. System Functionalities

4.1. Description: The system operates as a stateful, AI-driven workflow engine that continuously monitors inventory and processes procurement activities through multiple interconnected stages. It leverages historical sales data and contextual information to predict demand and dynamically adjust reorder thresholds. Once a restocking need is identified, the system initiates an automated workflow that includes supplier communication, negotiation, invoice processing, and financial reconciliation, all while maintaining a clear and traceable workflow state throughout the process.

4.2. Key Functionalities:

- **Predictive Inventory Management**
  AI analyzes sales trends, product thresholds and current product stock to determine optimal restocking thresholds dynamically.

- **Batch Restocking Optimization**
  Identifies related products from the same supplier that are near their thresholds, enabling cost-efficient bulk restocking.

- **Autonomous Supplier Communication & Negotiation**
  Automatically generates supplier messages, interprets multilingual responses, and performs negotiation within predefined constraints.

- **Unstructured Input Understanding**
  Processes and interprets messy inputs such as text messages, PDFs, images, and voice notes, converting them into structured procurement data.

- **Intelligent Invoice Processing**
  Extracts key invoice details and validates them against negotiated terms to ensure accuracy and detect inconsistencies.

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- **Financial Reconciliation & FX Conversion**
  Performs currency conversion and generates payment-ready summaries for final settlement.

- **Stateful Workflow Tracking**
  Maintains clear workflow states such as low stock detection, negotiation progress, invoice validation, and approval to ensure transparency and traceability.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: The system utilizes the Z.AI GLM model as its core intelligence layer due to its strong capabilities in reasoning, structured output generation, and handling unstructured inputs. This model is essential for enabling multi-step decision-making processes such as demand prediction, negotiation strategy, and invoice interpretation.

4.3.2. **Prompting Strategy:** A multi-step agentic prompting approach is adopted, where the backend preprocesses relevant data before passing structured context to the model. The model is then prompted to perform specific reasoning tasks and return structured outputs, typically in JSON format. This approach improves reliability, reduces hallucination, and ensures that outputs can be seamlessly integrated into the workflow system.

4.3.3. **Context & Input Handling**: The system is designed to handle unstructured and large inputs by applying preprocessing techniques such as chunking and summarization. Inputs that exceed the model's context limits are either truncated or processed incrementally. This ensures that the model receives relevant and manageable data while maintaining overall system performance.

4.3.4. **Fallback & Failure Behavior:** In cases where the model produces off-topic, incomplete, or unreliable responses, the system employs fallback mechanisms such as retrying with adjusted prompts, reverting to rule-based logic, or escalating the issue to the user. This ensures robustness and maintains trust in the system's outputs.

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5. User Stories & Use Cases

- **User Stories:** *As a merchant, I want the system to automatically detect when inventory is low so that I do not need to manually monitor stock levels. As a merchant, I want the system to handle supplier communication and negotiation so that I can save time and achieve better pricing outcomes. As a merchant, I want invoices to be automatically processed and validated so that I can ensure financial accuracy without manual effort. As a merchant, I want to track the status of each procurement workflow so that I always know what stage the process is in.*
- **Use Cases (Main Interactions):**

  Restocking Flow
  - System detects low stock based on AI-adjusted thresholds
  - Automatically initiates procurement workflow

  Negotiation Flow
  - AI generates supplier messages
  - Interprets supplier responses
  - Decides to accept, counter-offer, or escalate based on constraints

  Invoice Processing Flow
  - System extracts invoice data from uploaded documents
  - Validates data against negotiated terms

  Approval Flow
  - System generates transaction summary
  - User reviews and approves before finalization

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

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6. Features Included (Scope Definition)

- The system includes a fully automated procurement workflow that integrates inventory prediction, supplier communication, negotiation, invoice processing, and financial reconciliation into a single platform. It also provides a centralized dashboard for tracking workflow states and incorporates a human-in-the-loop mechanism for critical decision points.

7. Features Not Included (Scope Control)

- The system does not include full integration with real payment gateways or external supplier systems, as these are beyond the scope of the current implementation. It also does not support multi-user collaboration or production-level deployment, focusing instead on demonstrating the core AI-driven workflow capabilities.

8. Assumptions & Constraints

- **LLM Cost Constraint:**
  The system assumes that LLM usage costs will be managed through backend preprocessing and efficient prompt design. By reducing unnecessary token usage and caching repeated queries, the system aims to keep inference costs manageable for each user session.

- **Technical Constraints:**
  The system depends on the availability and performance of the Z.AI API and requires a workflow orchestration layer for managing stateful execution. Integration with real-world supplier systems is limited in this prototype.

- **Performance Constraints:**
  Processing large volumes of unstructured data may introduce latency, particularly during AI inference. While optimization techniques are applied, performance may vary depending on input complexity.

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- **User Input:**
  The system assumes that users will provide necessary parameters such as acceptable price ranges and supplier details. Human input is also required for final approval and in cases where escalation occurs.

9. Risks & Questions Throughout Development

- One of the key risks is the accuracy of AI-generated decisions, particularly in negotiation and invoice validation scenarios. Misinterpretation of supplier inputs or incorrect reasoning could lead to suboptimal outcomes. Another concern is handling ambiguous or incomplete unstructured data, which may require additional user clarification. System reliability is also a potential risk, especially when external APIs fail or return unexpected results. Additionally, there is an open question regarding how well the system can generalize across different industries, as procurement workflows may vary significantly depending on business context.

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